PROJECT REPORT

Design and Implementation of 8-bit Adder/Subtractor using Vivado Software

Submitted by

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1 Aim

The aim of this project is to design and implement an efficient 8-bit adder/subtractor circuit using Verilog Hardware Description Language (HDL) and simulate its functionality using Vivado Design Suite. The project focuses on developing a digital system capable of performing both addition and subtraction operations on two 8-bit binary numbers .

2 Introduction

Digital electronics plays a vital role in modern technology, forming the foundation of computers, communication systems, and embedded devices. Arithmetic operations such as addition and subtraction are among the most fundamental processes performed in digital systems. These operations are essential in applications ranging from simple calculators to complex processors and digital signal processing systems. Therefore, designing efficient arithmetic circuits is a key aspect of digital system design.

An 8-bit adder/subtractor is a combinational logic circuit that performs both addition and subtraction of two 8-bit binary numbers. Instead of designing separate circuits for addition and subtraction, a single unified circuit can be implemented using the concept of 2's complement arithmetic. In this method, subtraction is carried out by taking the 2's complement of one operand and adding it to the other operand. A control signal is used to determine whether the circuit performs addition or subtraction, making the design more efficient and reducing hardware complexity.

With the advancement of Electronic Design Automation (EDA) tools, designing and testing digital circuits has become more efficient and accurate. In this project, the design and simulation of the 8-bit adder/subtractor are carried out using the Vivado Design Suite. Vivado provides a comprehensive environment for writing Verilog code, performing functional simulation, synthesizing the design, and implementing it on FPGA platforms. It allows designers to verify the correctness of the circuit before hardware implementation.

3 Theory

Digital arithmetic circuits are essential components in modern electronic systems. The design of an 8-bit adder/subtractor is based on fundamental building blocks such as half adders, full adders, and the concept of 2's complement arithmetic.

3.1 Full Adder

A full adder is an extension of the half adder that includes a carry input along with the two input bits. It is capable of performing the addition of three bits: A, B, and C_{in}. The output equations are:

$$S = A \oplus B \oplus C_{in} \tag{1}$$

$$C_{out} = AB + BC_{in} + AC_{in} \tag{2}$$

Full adders are used as the basic building blocks to construct multi-bit adders.

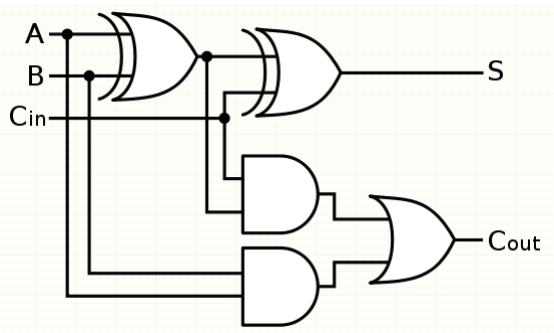

Figure 1: Full Adder Circuit

3.2 Ripple Carry Adder

An 8-bit adder is implemented by cascading eight full adders, where the carry output of one stage is connected to the carry input of the next stage. This type of adder is known as a ripple carry adder because the carry propagates through each stage sequentially.

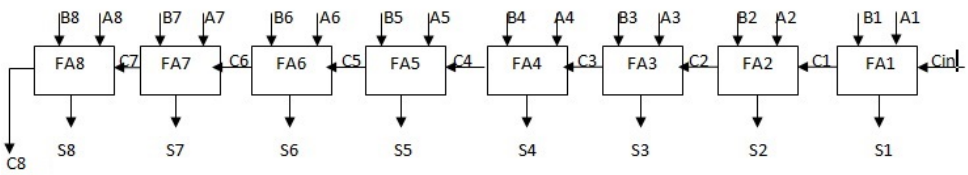

Figure 2: Ripple Carry Adder

However, the propagation delay increases with the number of bits due to sequential carry propagation.

3.3 Adder/Subtractor Operation

An adder/subtractor circuit combines both addition and subtraction operations using a control signal M. The modified input is given by:

$$B' = B \oplus M \tag{3}$$

The final output becomes:

$$Result = A + B' + M \tag{4}$$

When $M = 0$, the circuit performs addition:

$$Result = A + B$$

When $M = 1$, the circuit performs subtraction:

$$Result = A + \overline{B} + 1 = A - B$$

Thus, by using XOR gates and a control signal, a single circuit can efficiently perform both addition and subtraction operations.

4 Circuit Diagram

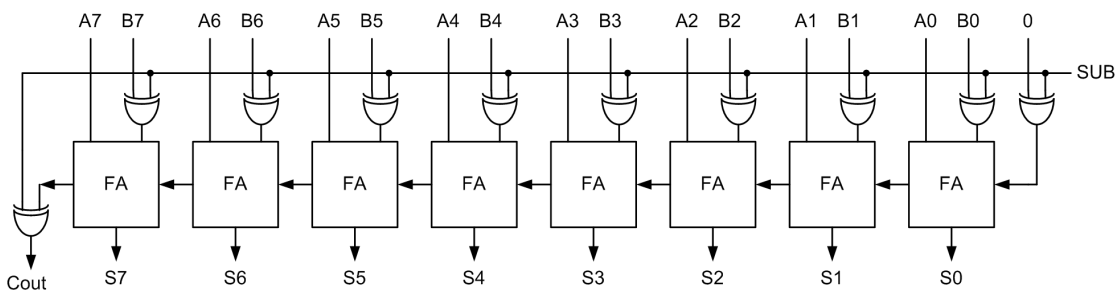

Figure 3: 8-bit Adder/Subtractor Circuit

5 Program Code

5.1 Verilog Code for 8-bit Adder/Subtractor

```
module adder_subtractor_8bit (
    input [7:0] A,
    input [7:0] B,
    input M,
    output [7:0] S,
    output Cout
);

    wire [7:0] w;
    wire [6:0] carry;
    wire cin;


    xor x0(w[0],M,B[0]);
    xor x1(w[1],M,B[1]);
    xor x2(w[2],M,B[2]);
    xor x3(w[3],M,B[3]);
    xor x4(w[4],M,B[4]);
    xor x5(w[5],M,B[5]);
    xor x6(w[6],M,B[6]);
    xor x7(w[7],M,B[7]);


    full_adder FA0 (A[0],w[0], M,S[0],carry[0]);
    full_adder FA0 (A[1],w[1], carry[0],S[1],carry[1]);
    full_adder FA0 (A[2],w[2], carry[1],S[2],carry[2]);
    full_adder FA0 (A[3],w[3], carry[2],S[3],carry[3]);
    full_adder FA0 (A[4],w[4], carry[3],S[4],carry[4]);
    full_adder FA0 (A[5],w[5], carry[4],S[5],carry[5]);
    full_adder FA0 (A[6],w[6], carry[5],S[6],carry[6]);
    full_adder FA0 (A[7],w[7], carry[6],S[7],carry[7]);

endmodule
```

```
// Full Adder Module
module full_adder (
    input A, B, Cin,
    output Sum, Cout
);

assign Sum = A ^ B ^ Cin;
assign Cout = (A & B) | (B & Cin) | (A & Cin);

endmodule
```

5.2 Testbench Code

```
`timescale 1ns/1ps

module tb_adder_subtractor;

    reg [7:0] A, B;
    reg M;
    wire [7:0] S;
    wire Cout;

    // Instantiate DUT
    adder_subtractor_8bit uut (
        .A(A),
        .B(B),
        .M(M),
        .S(S),
        .Cout(Cout)
    );

    initial begin
        $display("Time\tA\tB\tM\t|\tS\tCout");
        $monitor("%0t\t%d\t%d\t%b\t|\t%d\t%b", $time, A, B, M, S, Cout);

        // ADDITION TESTS
        A = 8'b11011010; B = 8'b10001111;  M = 0; #10;
```

```
        A = 8'b10010010; B = 8'b11111111; M = 0; #10;

        // SUBTRACTION TESTS
       A = 8'b11011010; B = 8'b10101110;  M = 1; #10;
        A = 8'b10010010; B = 8'b11111111; M = 1; #10;
        $finish;
    end

endmodule
```

6 Simulation Results

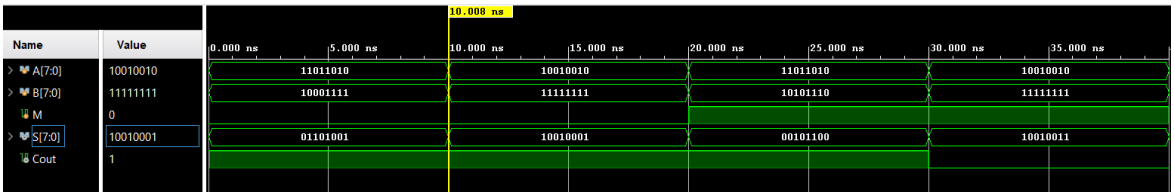

Figure 4: Simulation Result

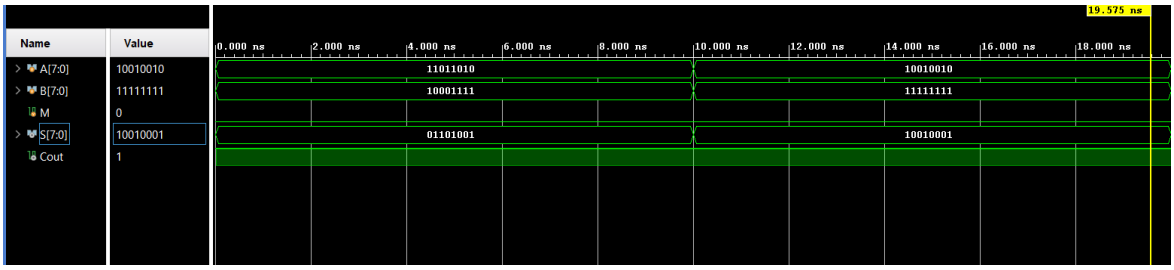

Figure 5: Addition Result

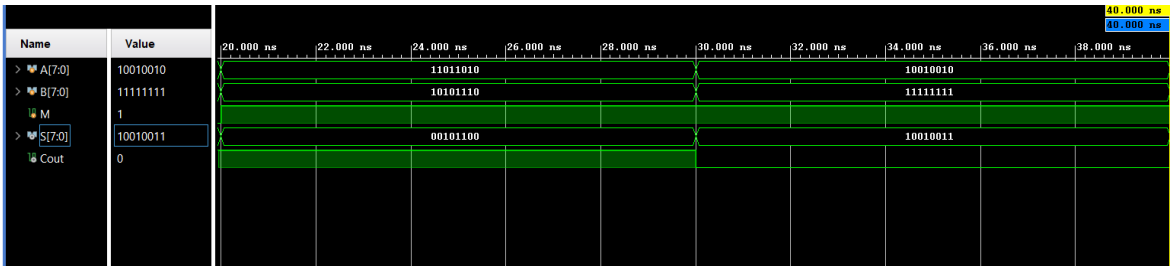

Figure 6: Subtraction result

7 Result and Discussion

The functionality of the circuit was verified by applying various test cases through the testbench, and the corresponding output waveforms were observed. For the addition operation (M = 0), the circuit correctly performed binary addition of the two 8-bit inputs A and B. The output results matched the expected sum values, and the carry-out (Cout) was generated appropriately whenever an overflow occurred. For the subtraction operation (M = 1), the circuit successfully implemented subtraction using the 2's complement method. The input B was inverted and an additional 1 was added through the carry input, enabling accurate subtraction. The results obtained from simulation matched the expected values.

8 Conclusion

In this project, an 8-bit adder/subtractor was successfully designed and implemented using Verilog HDL and simulated using the Vivado Design Suite. The circuit was able to perform both addition and subtraction operations efficiently using a single hardware structure by employing the concept of 2's complement arithmetic.

The use of a control signal enabled seamless switching between addition and subtraction modes, thereby reducing hardware complexity and improving design efficiency. The simulation results verified that the circuit produced accurate outputs for all tested input combinations.

9 References

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3. Xilinx Inc., *Vivado Design Suite User Guide*, Available online: `https://www.xilinx.com`
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5. IEEE Standard for Verilog Hardware Description Language, IEEE Std 1364.